WORKED SOLUTION

STEP III 2019, Question 6

Pure • Circles in the Argand Diagram

STEP III 2019, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The opening expression is $|z-a|^2 - r^2$ in disguise, so the locus is the set of points at distance r from a. For the image under $\omega = 1/z$, substitute $z = 1/\omega$, clear the fractions, and then rearrange the result back into the *same* shape as the original equation. Once it's in that shape, we can read the centre and radius straight off by matching coefficients. Comparing centres and radii then settles when C' and C are the same circle.

Solution

Since $(z-a)^* = z^* - a^*$,

$$|z-a|^2 = (z-a)(z^*-a^*) = zz^* - a^*z - az^* + aa^*,$$

so the given equation says exactly that $|z-a|^2 = r^2$, that is, $|z-a| = r$ (as $r > 0$). Hence C is the circle with **centre** a **and radius** r.

More generally, an equation $ww^* - bw^* - b^*w + bb^* - s^2 = 0$ with $s > 0$ is the circle with centre b and radius s. We'll use this as a template.

Part (i)

First, $z = 0$ is not on C: that would need $|a| = r$, i.e. $a^*a = r^2$, which is excluded. So $\omega = 1/z$ is defined for every point of C, and $\omega \neq 0$. Also $z = 1/\omega$, so Q lies on C' exactly when $\omega \neq 0$ and

$$\left|\frac{1}{\omega} - a\right| = r \iff |1 - a\omega| = r|\omega| \iff (1-a\omega)(1-a^*\omega^*) = r^2\omega\omega^*.$$

Expanding and collecting terms,

$$(a^*a - r^2)\,\omega\omega^* - a\omega - a^*\omega^* + 1 = 0.$$

Write $k = a^*a - r^2 = |a|^2 - r^2$, which is a non-zero real number by hypothesis. Dividing by k,

$$\omega\omega^* - \frac{a^*}{k}\,\omega^* - \frac{a}{k}\,\omega + \frac{1}{k} = 0.$$

This matches the template with $b = a^*/k$ (so $b^* = a/k$, since k is real), provided $bb^* - s^2 = 1/k$, i.e.

$$s^2 = \frac{|a|^2}{k^2} - \frac{1}{k} = \frac{|a|^2 - k}{k^2} = \frac{r^2}{k^2}.$$

This is positive, so C' is a genuine circle, with

$$\text{centre } \frac{a^*}{|a|^2 - r^2}, \qquad \text{radius } \frac{r}{||a|^2 - r^2|}.$$

(Note that $\omega = 0$ doesn't satisfy the equation, since it would give $1 = 0$. So every point of this circle really is $1/z$ for some z on C.)

Ask yourself Two circles are the same exactly when…?

They are the same exactly when they have the same centre and the same radius. So $C' = C$ requires

$$\frac{r}{|k|} = r \qquad \text{and} \qquad \frac{a^*}{k} = a.$$

The first gives $|k| = 1$, i.e. $k^2 = 1$:

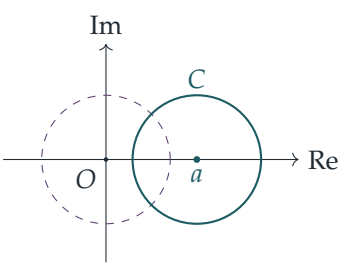
$$(|a|^2 - r^2)^2 = 1.$$

So $k = 1$ or $k = -1$, and then the second condition gives:

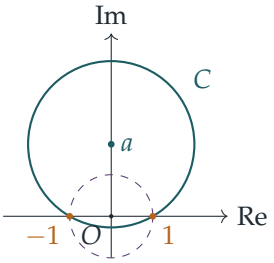
- if $k = 1$: $a^* = a$, so a is **real**;
- if $k = -1$: $a^* = -a$, so a is **imaginary** (purely imaginary).

(If $a = 0$ it is both, and then $r = 1$.) Conversely, either of these, together with the radius condition, does make the centres and radii agree.

Sketches. If a is real, then $|a|^2 = r^2 + 1 > r^2$, so the centre is on the real axis at distance $\sqrt{r^2+1}$ from O, and the origin lies *outside* C. The circle meets the real axis at $a - r$ and $a + r$, whose product is $a^2 - r^2 = 1$, as it must be, since $1/z$ swaps them. If a is imaginary, then $|a|^2 = r^2 - 1$, so this case needs $r \geq 1$. The centre is on the imaginary axis and the origin lies *inside* C. Writing $a = \mathrm{i}c$, the real-axis crossings satisfy $x^2 + c^2 = r^2$, so $x = \pm 1$. In both cases C crosses the unit circle (dashed) at right angles, because $|a|^2 = r^2 \pm 1$ (the left picture has $a = \sqrt{2}$, $r = 1$; the right has $a = \mathrm{i}\sqrt{3}$, $r = 2$).



a real



a imaginary

If a is real it may of course be negative, giving the mirror image of the left picture in the imaginary axis; similarly, an imaginary a may lie below the real axis.

Part (ii)

Now $z = 1/\omega^*$, so Q is on the locus exactly when

$$\left|\frac{1}{\omega^*} - a\right| = r \iff |1 - a\omega^*| = r|\omega^*| \iff |1 - a^*\omega| = r|\omega|,$$

taking the conjugate inside the first modulus (a number and its conjugate have the same modulus). This is the equation of part (i) with a replaced by a^*. So, by part (i), the locus is the circle with

$$\text{centre } \frac{a}{k}, \qquad \text{radius } \frac{r}{|k|}, \qquad k = |a|^2 - r^2.$$

For this to be C we need $|k| = 1$ and $a/k = a$. If $a \neq 0$, the second condition forces $k = 1$, and that's all: the conditions are exactly

$$|a|^2 = r^2 + 1,$$

which says nothing about the argument of a. (If $a = 0$ we need $r = 1$, as before.)

Ask yourself Can you find a specific a that is neither real nor imaginary?

Take $a = 1 + \mathrm{i}$ and $r = 1$. Then $|a|^2 = 2 = r^2 + 1$, so the locus of Q is C, but a is neither real nor purely imaginary. So the answer is **no**: it is not the case that a must be real or imaginary.

Checks

- For $a = \sqrt{5}$, $r = 2$ (so $|a|^2 - r^2 = 1$), the points $z = \sqrt{5} \pm 2$ on C go to $1/(\sqrt{5} \pm 2) = \sqrt{5} \mp 2$, which are again on C.
- In part (ii) with $a = 1 + \mathrm{i}$, $r = 1$: the point $z = 1$ is on C, and $1/z^* = 1$ is too. The point $z = 2 + \mathrm{i}$ gives $1/z^* = (2 + \mathrm{i})/5 = 0.4 + 0.2\mathrm{i}$, and $|(0.4 + 0.2\mathrm{i}) - (1 + \mathrm{i})|^2 = 0.36 + 0.64 = 1$. But $1/z = 0.4 - 0.2\mathrm{i}$ is not on C, so this circle is not fixed by the map in part (i).

- The image circles in both parts were also confirmed by computer, by mapping many points of random circles and measuring their distances from the predicted centres.

Common slips

- Writing the centre of C' as $a/(|a|^2 - r^2)$ instead of $a^*/(|a|^2 - r^2)$. Reading off the centre needs the coefficient of ω^*, not of ω.
- Forgetting the modulus in the radius $r/\left||a|^2 - r^2\right|$. Without it, the case $|a| < r$ gives a negative radius.
- Answering "yes" in part (ii) by analogy with part (i), without noticing that the conjugate removes the condition $a^* = \pm a$.

Beyond STEP

The map $z \mapsto 1/z^*$ is *inversion in the unit circle*, and $z \mapsto 1/z$ is that inversion followed by reflection in the real axis. Both belong to the family of Möbius transformations, which always send circles and lines to circles and lines. Part (ii) is the classical fact that a circle is fixed by inversion exactly when it cuts the circle of inversion at right angles, i.e. $|a|^2 = r^2 + 1$ (or is the unit circle itself). Needham's *Visual Complex Analysis* develops these ideas beautifully, with pictures.

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